Feb 17 2026

Behavior Based Robotics Lecture

- Issue: Coordinating multiple behaviors at once

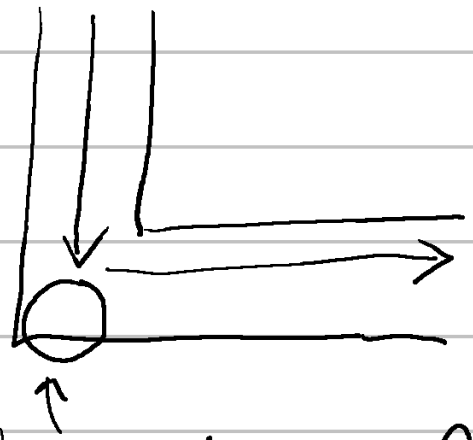
- Many functions will seek to manage the motors we need to have the bot manage these

Classical AI is about moving within an environment

Potential Stratagy for the maze

Follow the Right wall to the finish

- Potential issue we need to have the robot see the 90° wall

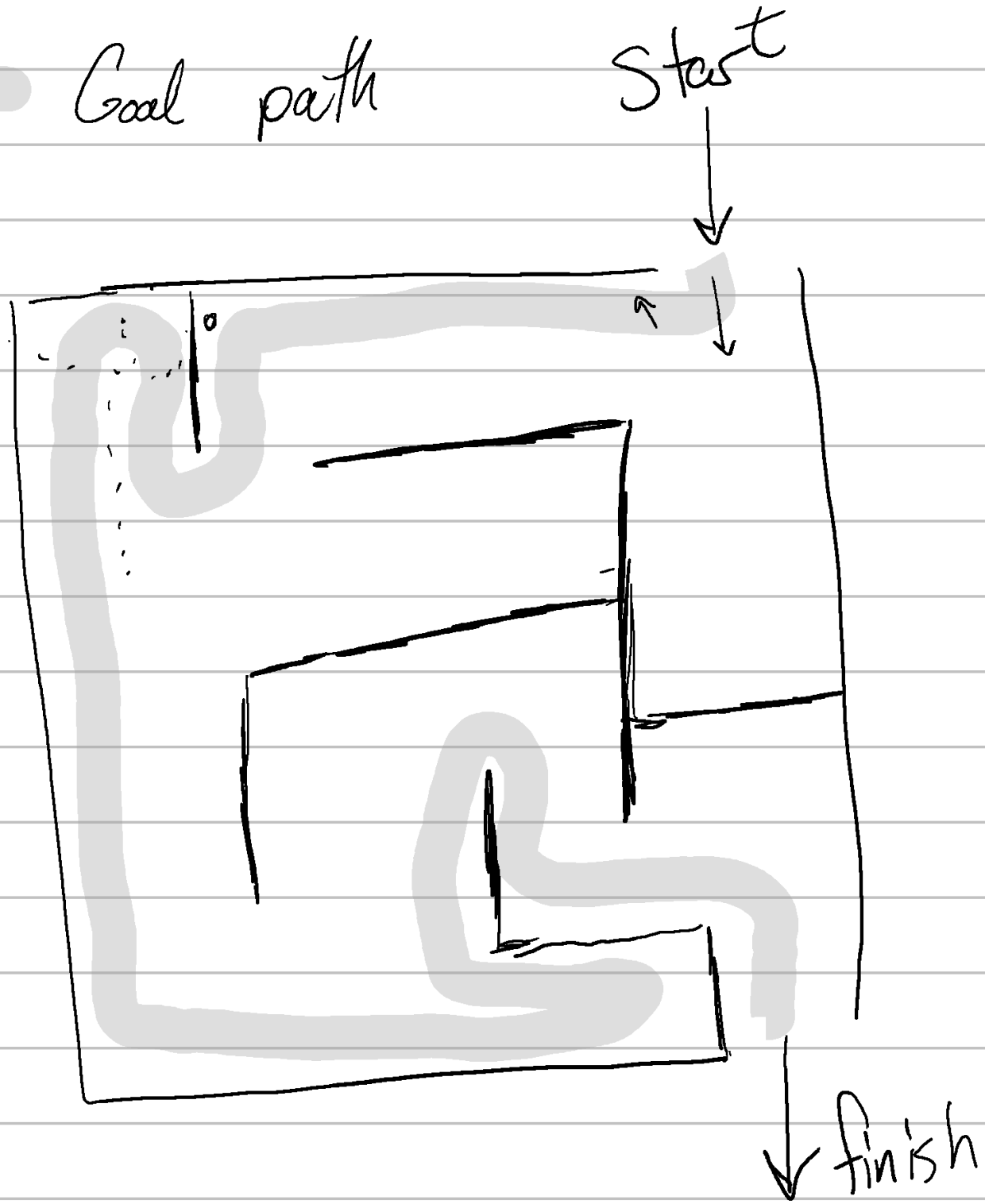


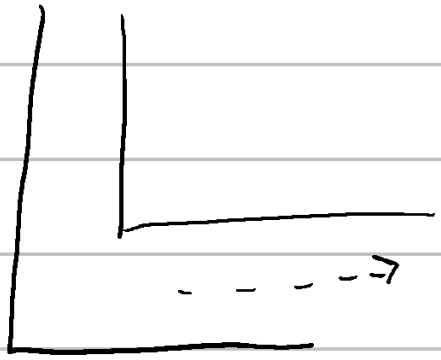
We need to figure out how the bot will efficiently see or bump into the wall.

- We may seek to have the bot maintain a set distance from the right wall

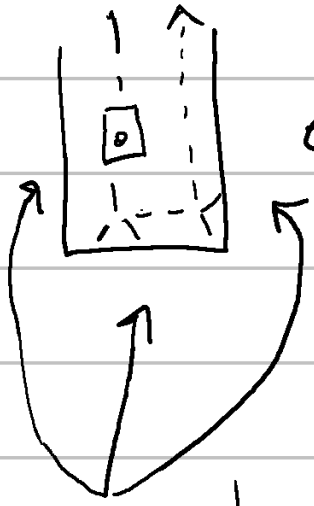
I will mark problem areas

Goal path

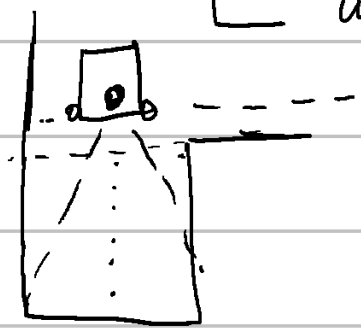




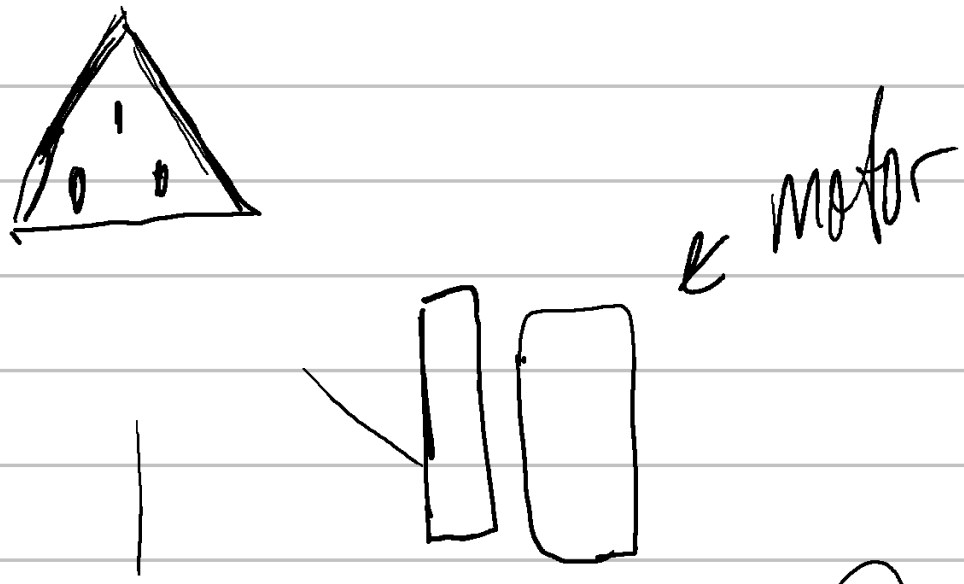
a obj sensor will see the lack of a left facing wall

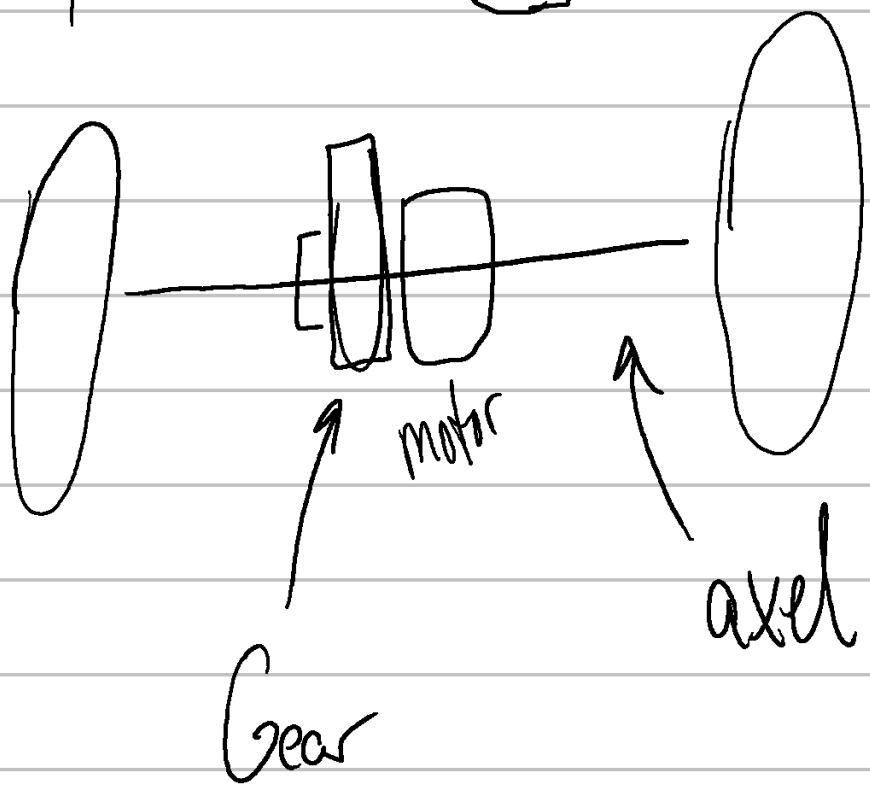


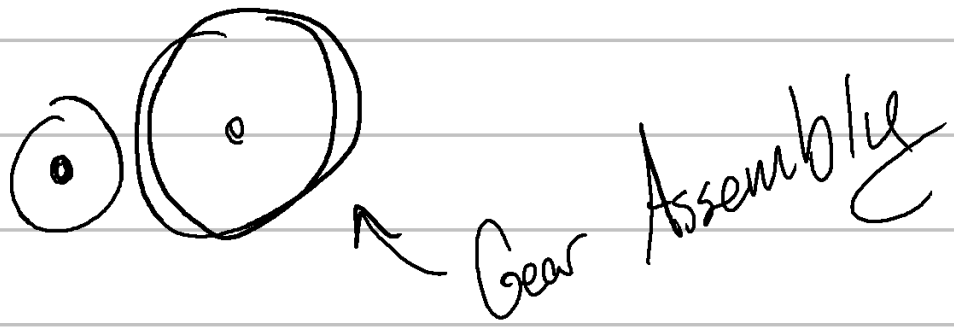
this sort of turn will be an issue in thinking we impliment a case where some sensor sees three walls and does 180° turns.



ideally we have a sensor see this wall is close and turn away. Making the bot detect that there is something on the right and that wall is x close distance and treat it as a Right turn


motor


motor
axel
Gear


Gear Assemblye

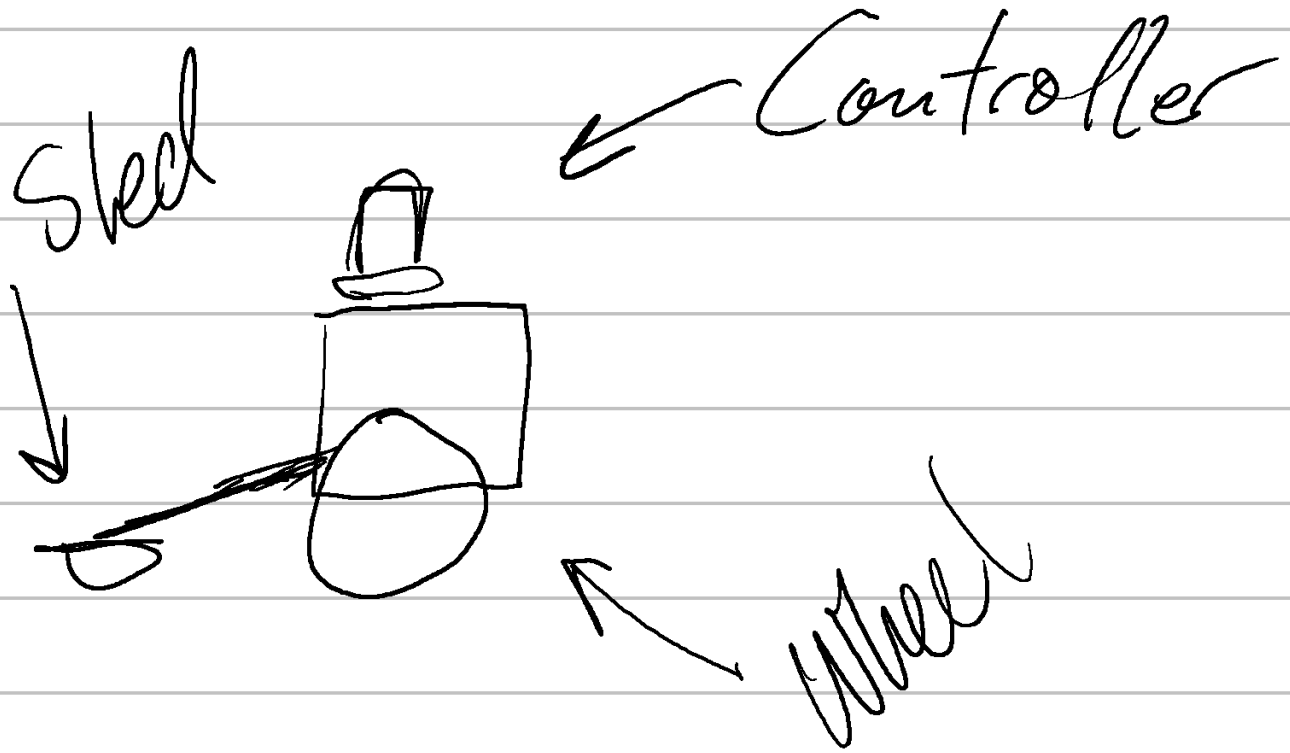
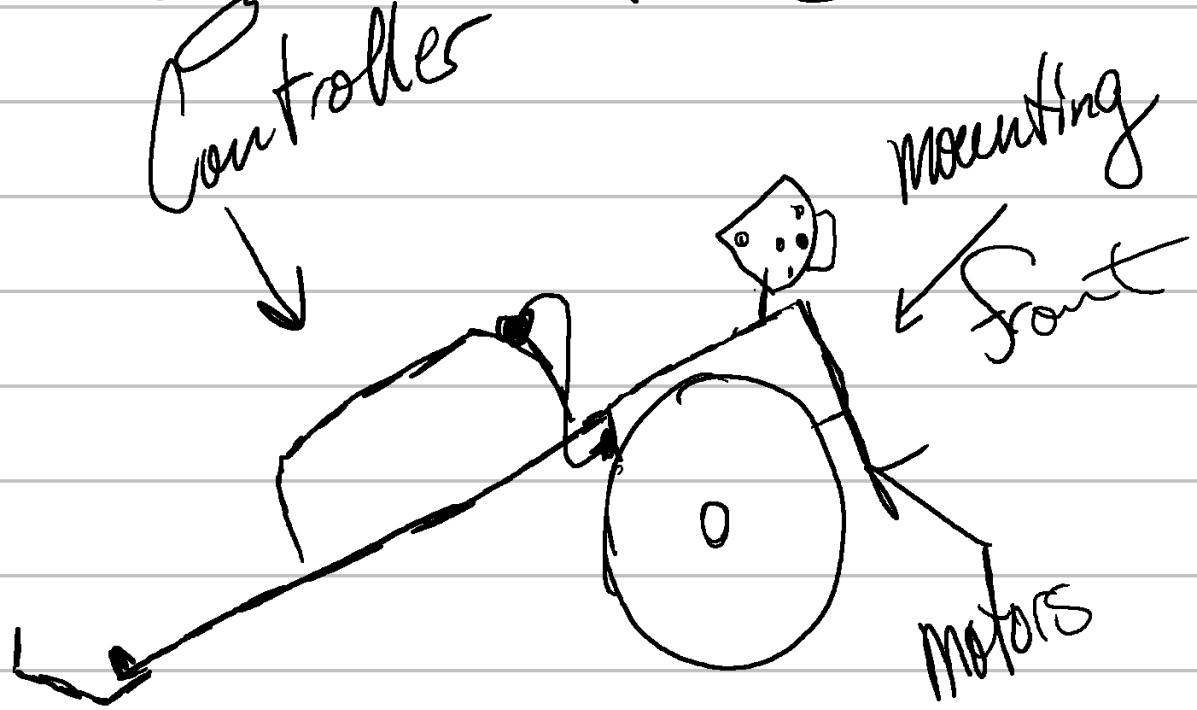
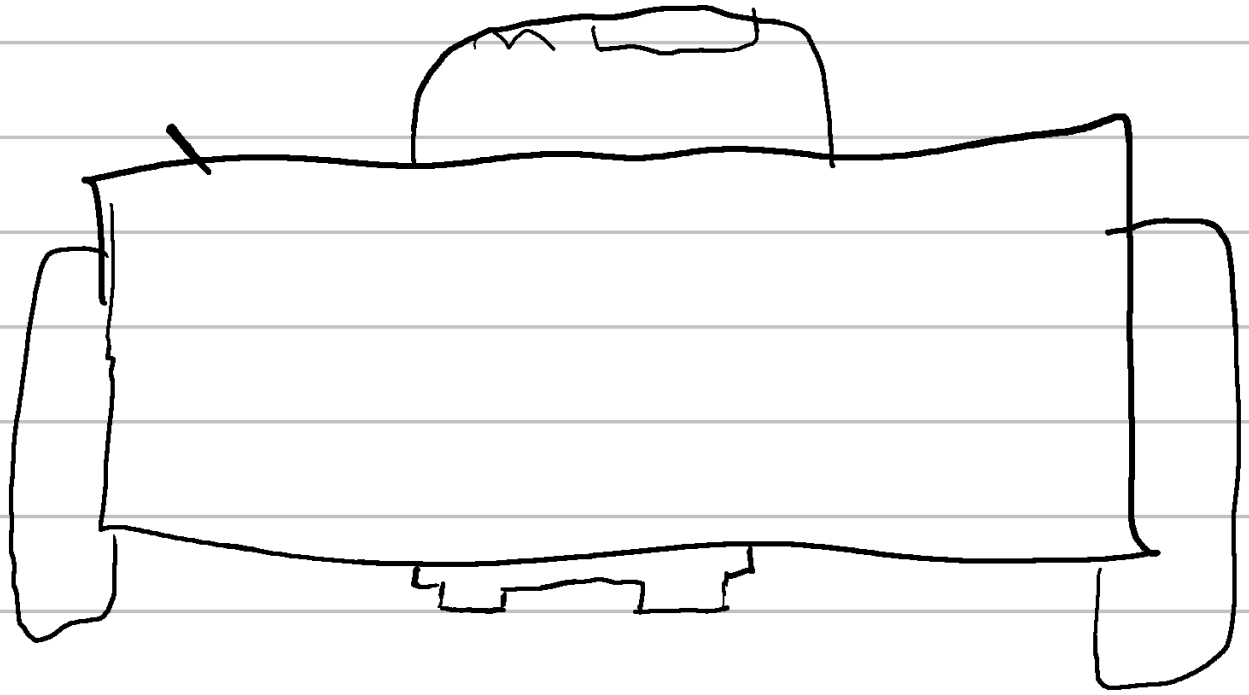
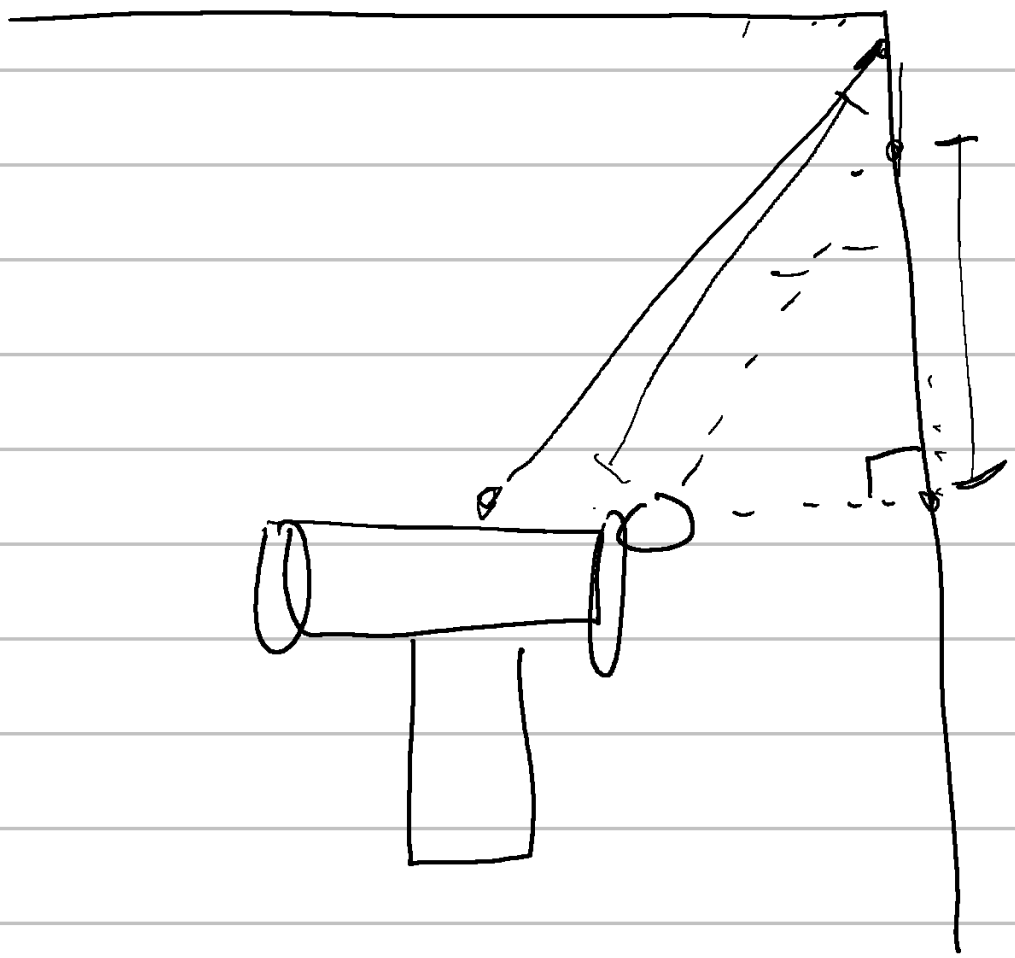

Controller
Sled
Wheel

Diagram of Sled Bot
"Teddy Rollsavelt"

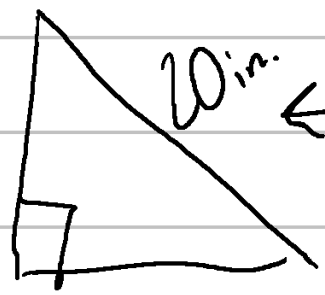


Goal in design size reduction





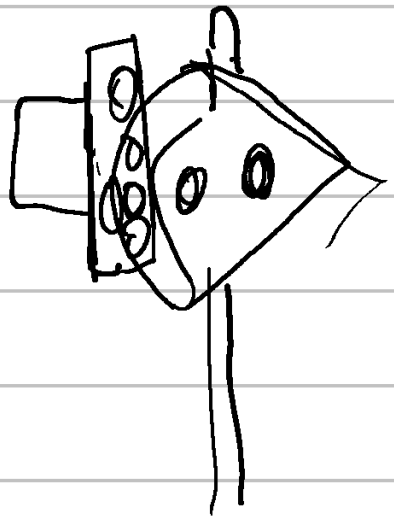
Potential Way to measure distance



We know this

In theory this is 90°

attempted Sketch



New thought we will mount a distance sensor to the front. So that the bot can dynamically see distance when needed.

We needed the ability with our angled Bot to change the angle which distance sensor is pointed. So I created a pinning system in which the mounts could be pinned in place by two screws.

2/24/20

- We began to work on our code

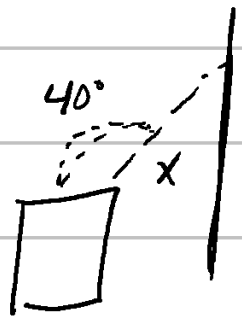
- We are talking a lot about functionality and how to make sure it always follows the wall

Testing Wall Following
It turn ✓
It detect wall ✓

2/26/26

We are attempting to impliment code that allow the bot to see a change in wall direction and turn.

Our biggest issue is we are struggling to get a measure in which we will work towards getting it to turn. So then what I am thinking we do is have the bot maintain a wall then



So the bot needs to maintain that x distance

So then when that wall is approaching the bot will see its set range has been jumped and turn

2/27/26

- We are still working on wall following.
- Whats wrong with wall following currently.
  - Senses the wall and reacts too violently
  - We currently have it sensing the wall however it hits the wall in random directions
- We need to have the bot straighten itself when the bot turns.
- We also need the bot to maintain its distance from the wall.
- We added a bool checker to makes sure the bot is actually getting to an alternate case and not running adjust infinitly.

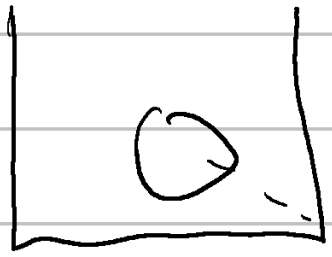
Adj inf did not improve moving away from light sensor too inconsistent

3/3/26

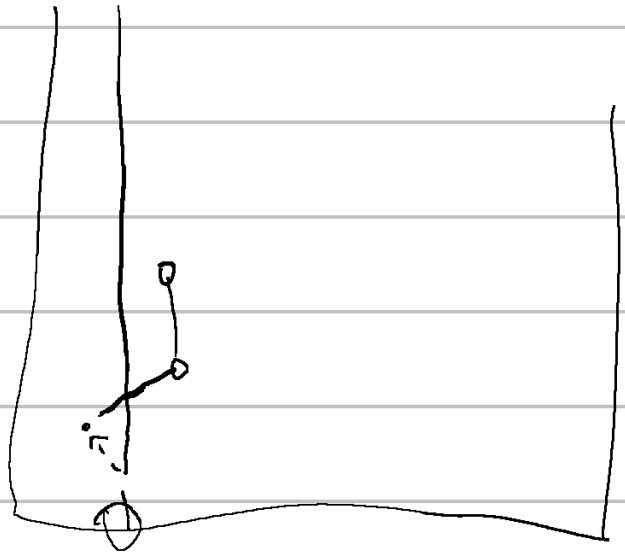
- Opened Code : Broke
- Fixed Code : Works

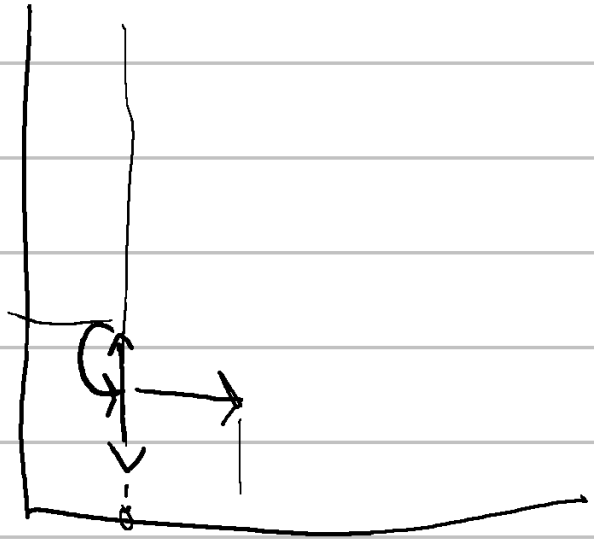
- Current issues : Bot does not maintain its tolerance all that well
- The bot will not set itself back to 8.0 from the wall. When the code sees its out of tolerance it will set itself to turn which is the wrong choice. When too close to the wall. The issue is in our tracer we have boolean checks which when true initates a turn.

· Issue of getting around a box persists



That falls outside tolerance





3/6/26

Found issue of wheels rubbing on mounting potentially causing issue of odd wheel velocities.

- Putting Spacers on the wheels did not solve this issue.
- Wheel motor doesn't output equally however. We already have adj. functionality.
- The big problems currently are the wheel axels and the tail. We are attempting to shorten the axels more we are attempting to code in a way for the bot to round corners...

We have majority of logic working ~~need~~ fine tuned

all

3/12/26

Again logic works However we still have the issue of getting caught on the wall so we removed the tain and replaced it w/ screw heads.

We turned up the velocity as well

Code Runs Slow...
- This is due the turn functionality being Slow and Jittery...

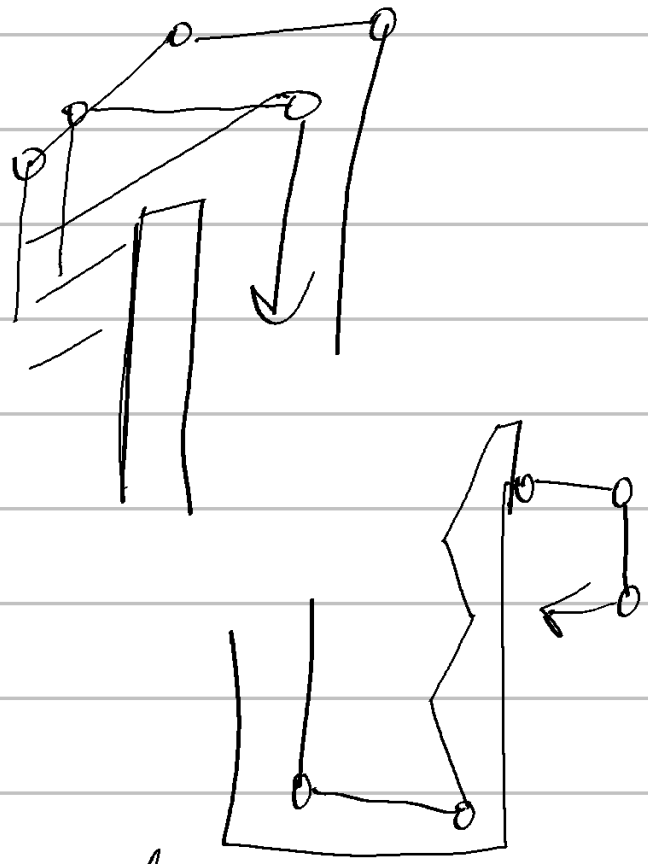
- Thus begins testing

- Group 3: They use much the same as They used a bump sensor it got caught a few times

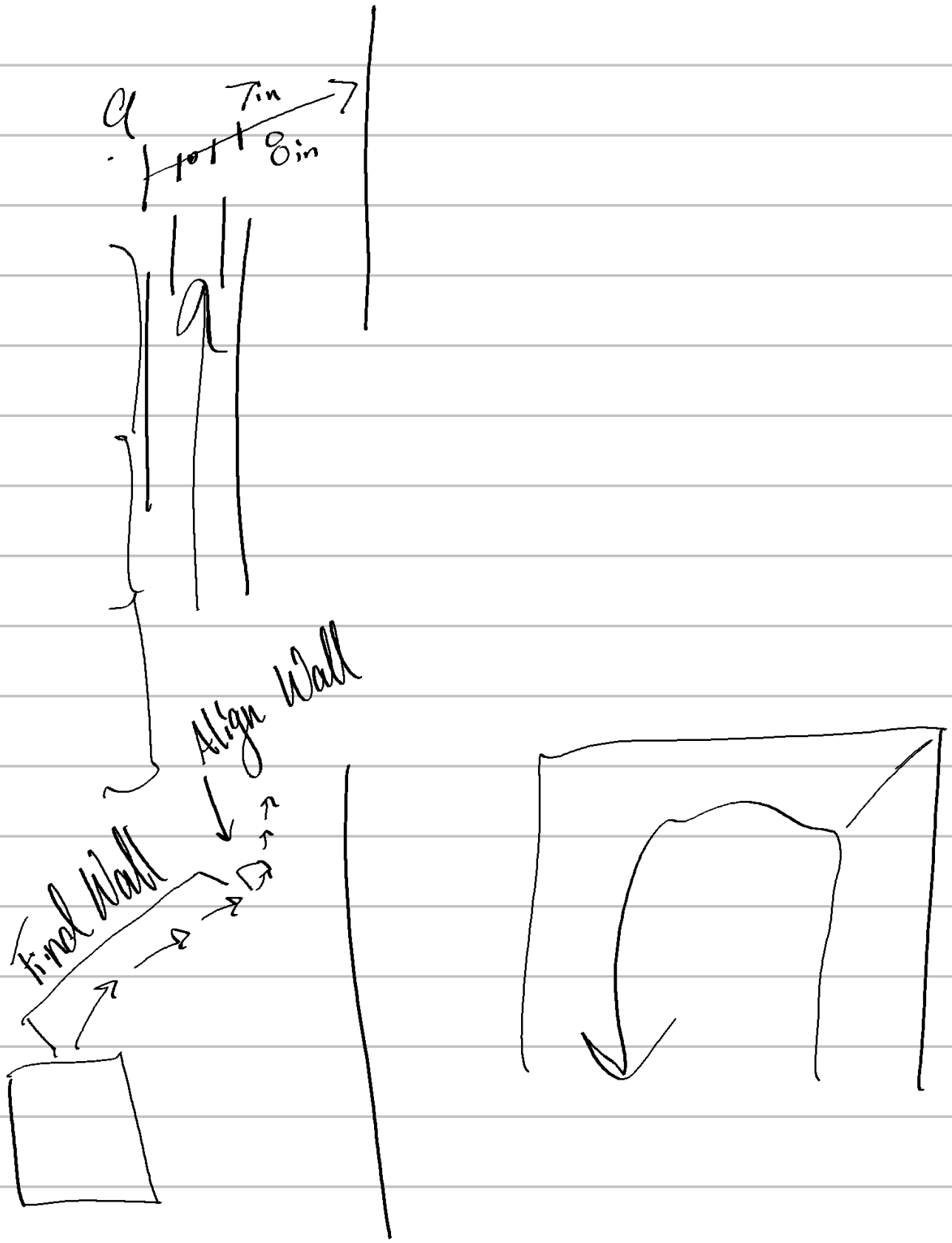
- Group 11 - Use the Hue Sensor as a "Bump Sensor"... Still bumps though
  - Does Run Super Smoothly
    - 1m 55sec.
- Group 10: They used a similar idea like our distance sensor but again using the Hue as a bump.

Group 5:
Set point on wall and turn pretty smart.

US: Problem:
- Bot tolerance functions failed us. Too much movement



Tolerance Got turned Down


Tin
Align Wall
Fixed Wall

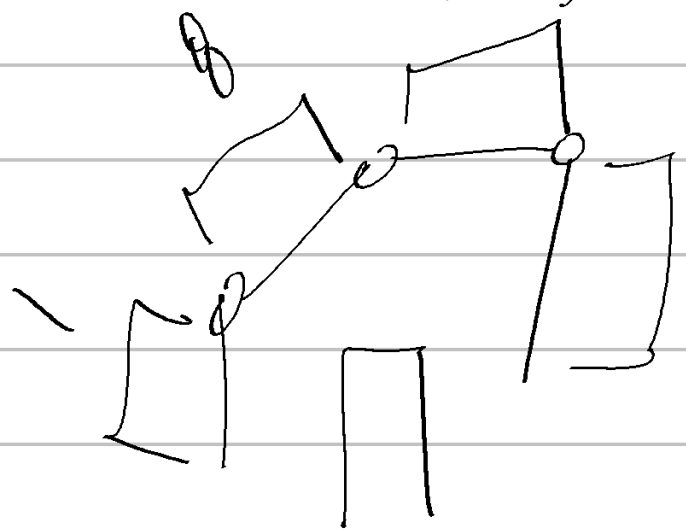
We turned up the tolerance from

Target: 7.25
Tolerance: 0.25

Target: 8
Tolerance: 1

Chicken Wrap Appears

10 from 8



Pseudocode

initialize ...

Set target Dist = 8 in

Set tolerance = 1 in

move forward

Find nearest wall

Align robot with wall

Loop forever

    Read distance

    Drive forward

    if distance > large gap threshold

        Stop robot

        move forward

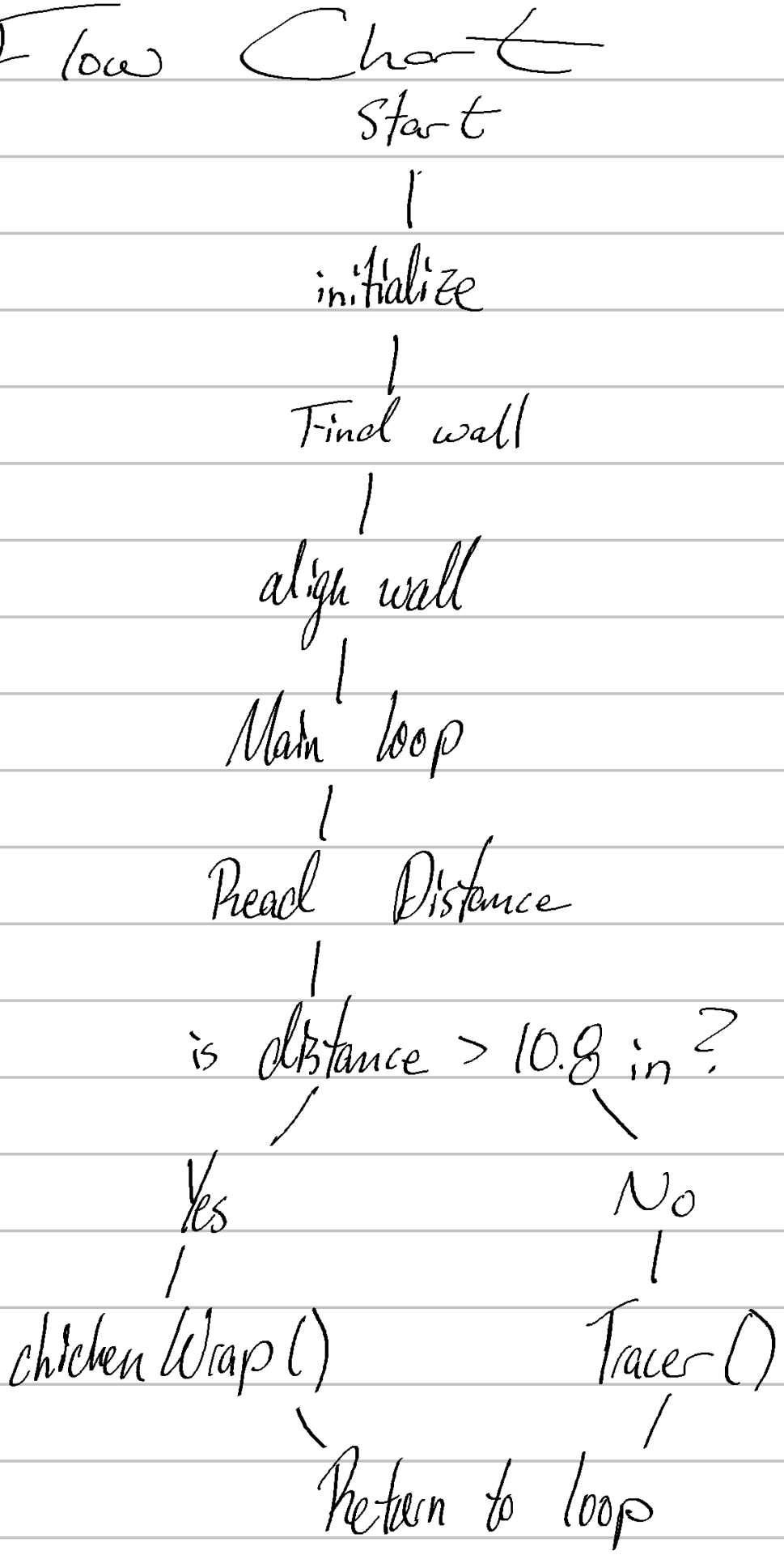
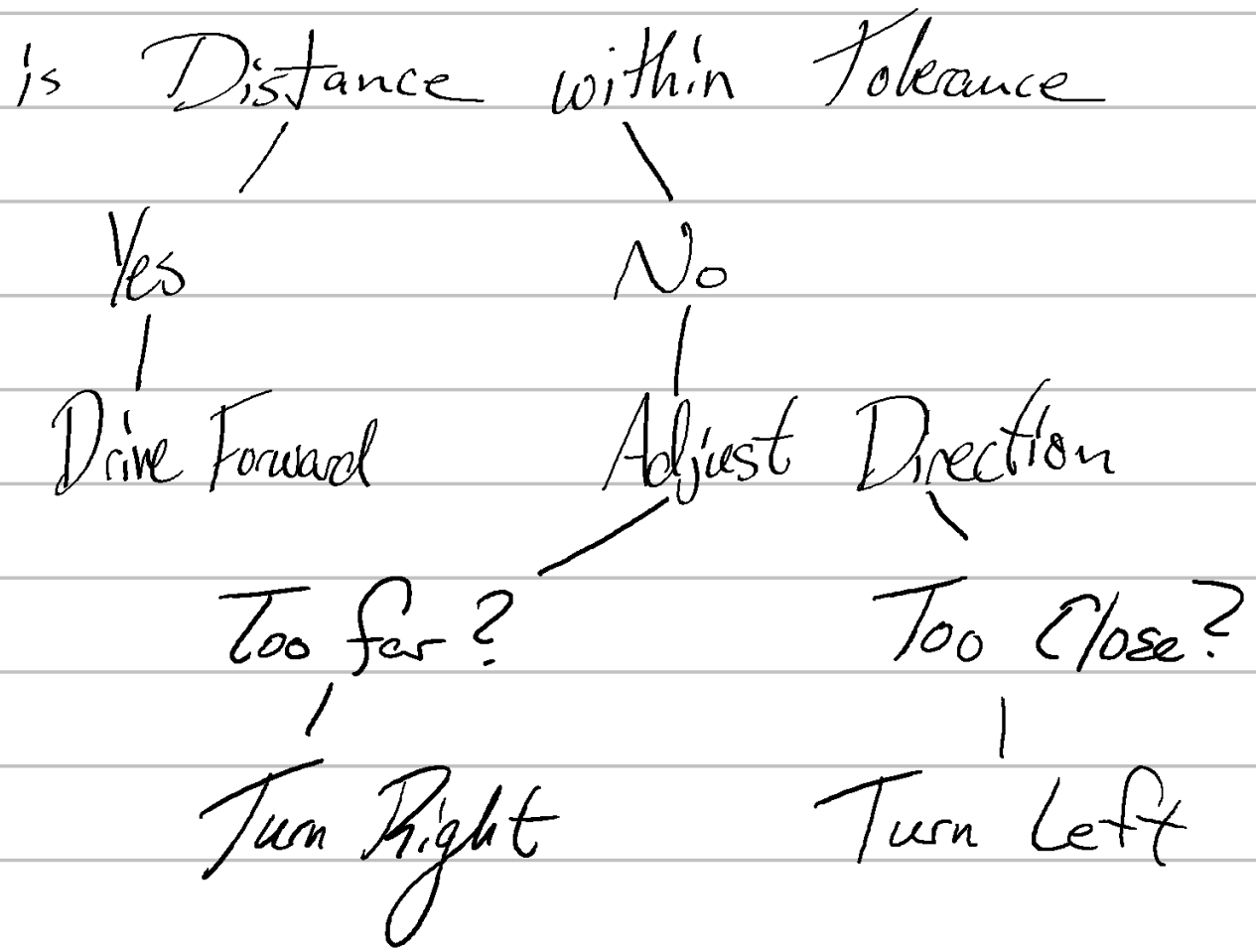
        perform corner navigation (chickenwrap)

Continued...

```
else
    perform wall tracing Adjust
        if too far from wall
            turn slightly towards wall
        elif too close
            turn slightly away from wall
        else
            continue forward

End of Loop
```

Flow Chart


Start
initialize
Find wall
align wall
Main loop
Read Distance
is distance > 10.8 in?
Yes
No
chickenWrap()
Tracer()
Return to loop

Tracer ( ) Flow Chart